This lemma is a slight variation of a similar result in [57], which does not impose any constraints on the determinant of 2-level unitaries (See also [1, 2]). For completeness, we present the proof in Appendix D.

Recall that in the decomposition $V = \bigoplus_{m=0}^{n} V^{(m)}$, unitary $V^{(m)}$ acts on the subspace with Hamming weight m, which has dimension $\binom{n}{m}$. Then, applying this lemma, we find that $V^{(m)}$ can be realized with

$$\frac{1}{2}\binom{n}{m} \times \left[\binom{n}{m} - 1\right]$$

2-level energy-conserving unitaries, each of which acts on two computational basis vectors in the subspace of Hamming weight m, has determinant 1, and can be constructed with the method in Step 2.

In conclusion, the total number of 2-level gates needed to implement V is

$$\frac{1}{2}\sum_{m=1}^{n-1}\binom{n}{m}\left[\binom{n}{m} - 1\right] = \frac{\binom{2n}{n} - 2^n}{2} \approx \frac{4^n}{2\sqrt{\pi n}}\,, \quad (77)$$

where $\approx$ means the ratio of two sides goes to 1, in the limit $n \to \infty$.

Combining this with Corollary 11 we conclude that

Proposition 13. *Any energy-conserving unitary in $\mathcal{SV}_n^{U(1)}$ can be realized with $\mathcal{O}(4^n n^{3/2})$ gates in any of the gate sets 1 and 2 in Theorem 1, without ancillary qubits.*

Here, to count the number of gates, one can use the diagram in Fig. 6: Recall that controlled-iSWAP can be realized using the circuit in Fig. 4, which requires $\mathcal{O}(1)$ gates in one of the aforementioned gate sets. Therefore, the total required number of elementary gates is

$$\left[\mathcal{O}(n^2) + \mathcal{O}(n)\right] \times \mathcal{O}\left(\frac{4^n}{\sqrt{n}}\right) = \mathcal{O}(4^n n^{3/2})\,. \quad (78)$$

Note that, in general, if the desired unitary acts non-trivially only on a subspace spanned by $D \leq 2^n$ basis elements, then from Lemma 12 it can be decomposed into no more than $D(D-1) = \mathcal{O}(D^2)$ 2-level unitaries. Hence, any such unitary can be implemented with $[\mathcal{O}(n^2) + \mathcal{O}(n)] \times \mathcal{O}(D^2) = \mathcal{O}(n^2 D^2)$ elementary gates in Theorem 1.

Step 4: Universality

Finally, applying a mechanism developed in [42], we show how one can achieve universality using a single ancillary qubit. It is worth noting that this technique can be applied to symmetric circuit with any arbitrary Abelian symmetry (See Lemma 5 of [42]).

Let $|\mathbf{b}\rangle$ be an arbitrary element of the computational basis $\{|0\rangle, |1\rangle\}^{\otimes n}$, other than $|0\rangle^{\otimes n}$. Then, there is (at least) one qubit with reduced state $|1\rangle$. Let $|\mathbf{b}'\rangle$ be the n-qubit state obtained from $|\mathbf{b}\rangle$ by changing the state of this qubit from $|1\rangle$ to $|0\rangle$. Define the Hamiltonian

$$\widetilde{H}_{\mathbf{b}} = |\mathbf{b}\rangle\langle\mathbf{b}| \otimes |0\rangle\langle 0|_{\text{anc}} - |\mathbf{b}'\rangle\langle\mathbf{b}'| \otimes |1\rangle\langle 1|_{\text{anc}}\,. \quad (79)$$

Note that $(n+1)$-qubit states $|\mathbf{b}\rangle|0\rangle_{\text{anc}}$ and $|\mathbf{b}'\rangle|1\rangle_{\text{anc}}$ have equal Hamming weights and their Hamming distance is 2. Furthermore, this Hamiltonian is traceless, which means $\exp(\mathrm{i}\widetilde{H}_{\mathbf{b}}\theta)$ has determinant 1. It follows from Step 1 that this 2-level energy-conserving unitary can be implemented without ancilla. Implementing this unitary on the system and ancilla we obtain state

$$\exp[\mathrm{i}\widetilde{H}_{\mathbf{b}}\theta](|\psi\rangle \otimes |0\rangle_{\text{anc}}) = \left(\exp[\mathrm{i}\theta|\mathbf{b}\rangle\langle\mathbf{b}|]|\psi\rangle\right) \otimes |0\rangle_{\text{anc}}\,, \quad (80)$$

where $|\psi\rangle \in (\mathbb{C}^2)^{\otimes n}$ is the initial state of n qubits in the system. Since the state of ancilla qubit remains unchanged, we can reuse again in a similar fashion. Therefore, in this way we can realize the unitary $\exp(\mathrm{i}|\mathbf{b}\rangle\langle\mathbf{b}|\theta)$. We can apply this procedure to any sector, except the sector with Hamming weight $m = 0$, which corresponds to state $|0\rangle^{\otimes n}$. It follows that using $(n-1)$ 2-level unitaries we can realize any unitary in the form

$$D = |0\rangle\langle 0|^{\otimes n} + \sum_{m=1}^{n} \exp(\mathrm{i}\theta_m)\,|\mathbf{b}_m\rangle\,\langle\mathbf{b}_m|\,, \quad (81)$$

for any arbitrary $\theta_1, \ldots, \theta_n \in (-\pi, \pi]$, where $|\mathbf{b}_m\rangle$ is an arbitrarily selected basis element in the sector with Hamming weight m.

Finally, recall that any energy-conserving unitary V can be decomposed as $V = e^{\mathrm{i}\theta_0} DQ$, where $e^{\mathrm{i}\theta_0} = V^{(0)} = \langle 0|^{\otimes n} V |0\rangle^{\otimes n}$, D is in the form of Eq. (81), with $\theta_m = \arg(\det(V^{(m)})) - \theta_0$ for $1 \leq m \leq n$, and $Q \in \mathcal{SV}_n^{\mathrm{U}(1)}$. As mentioned above, D can be implemented with an ancillary qubit. Furthermore, in the previous section we saw how $Q \in \mathcal{SV}_n^{\mathrm{U}(1)}$ can be implemented without ancillary qubits.

Note that the unitary generated by Hamiltonian in Eq. (79) is of the type we studied in Step 1 of this construction, and therefore can be realized with no more than $9(n+1)^2/8$ gates ($n+1$ qubits including one ancilla qubit) in $\mathcal{SV}_3^{\mathrm{U}(1)}$. In total, this step requires $\mathcal{O}(n^2)$ elementary gates.

In conclusion, any energy-conserving unitary can be realized, up to a global phase, with one ancillary qubit. Furthermore, using the diagram in Fig. 6 and Eq. (78), we find that this construction requires

$$\mathcal{O}(4^n n^{3/2}) + \mathcal{O}(n^2)(n-1) = \mathcal{O}(4^n n^{3/2})\,, \quad (82)$$

gates in one of these elementary gate sets, which completes the proof of Theorem 1 for sets 1 and 2.

VI. REALIZING ALL ENERGY-CONSERVING UNITARIES WITH XY INTERACTION ALONE

So far in this paper, we have considered circuits that contain both XY interaction as well as single-qubit rotations around the z axis. In this section, we focus on synthesizing quan-